

1.11 T2* DECAY (TRANSVERSE RELAXATION DUE TO FIELD INHOMOGENEITY)

T_2^* is the apparent transverse relaxation caused by both spin-spin interactions and magnetic field inhomogeneities.

T_2^* (T-two-star) is the time constant that describes the loss of transverse magnetization (M_{xy}) due to:

1. Microscopic spin-spin interactions (true T_2)
2. Macroscopic and microscopic magnetic field inhomogeneities (ΔB_0)

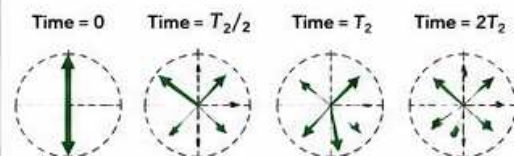
These local field variations cause spins to precess at slightly different frequencies, leading to faster dephasing and a shorter apparent decay than T_2 .

Relationship: $\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T_{2,inhomo}}$

WHY T_2^* IS SHORTER THAN T_2

A. Ideal (No Field Inhomogeneity) $\rightarrow T_2$

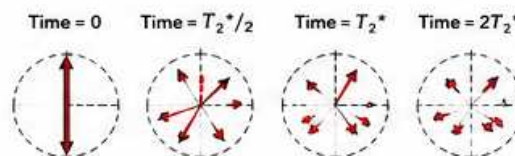
All spins precess at the same frequency.
Dephasing is due only to spin-spin interactions.



M_{xy} decreases with time constant T_2 .

B. With Field Inhomogeneity (ΔB_0) $\rightarrow T_2^*$

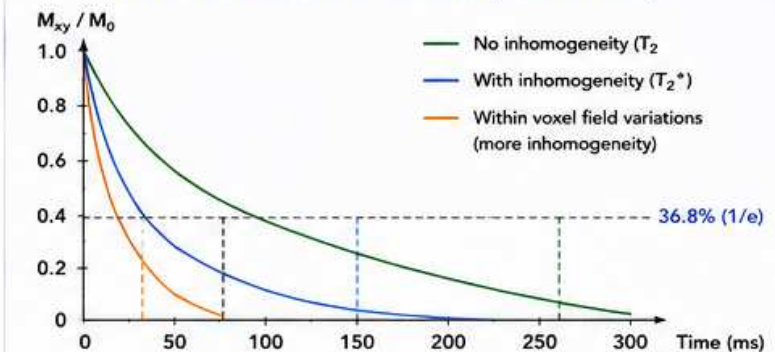
Spins precess at different frequencies.
Extra dephasing from field variations.



M_{xy} decays faster with time constant T_2^* .

T_2^* decay includes all dephasing mechanisms. Therefore, $T_2^* \leq T_2$ (always shorter or equal).

TRANSVERSE DECAY COMPARISON (After 90° Pulse)



More field inhomogeneity (ΔB_0) $\rightarrow$ faster decay $\rightarrow$ shorter T_2^* .
Better field homogeneity and shimming $\rightarrow$ longer T_2^* .

SOURCES OF FIELD INHOMOGENEITY (ΔB_0)

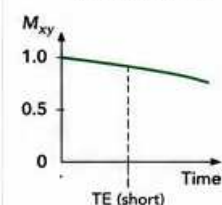
- B_0 main field inhomogeneity (imperfect magnet, fringe fields)
- Susceptibility differences (air-tissue, bone-tissue, metal, calcifications)
- Chemical shift (fat vs water)
- Eddy currents (rapidly switching gradients)
- Implanted metal / devices
- Patient motion

All cause spins within a voxel to see different local B_0 and precess at different frequencies.

EFFECT OF TE ON T_2^* WEIGHTING (Gradient Echo)

Short TE ($\ll T_2^*$)

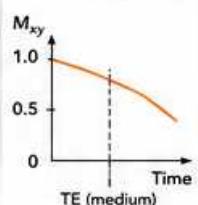
Little dephasing



Most signal remains.
Less T_2^* weighting.
Image is brighter.

TE $\approx T_2^*$

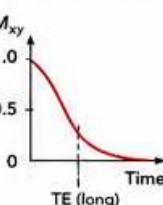
Moderate dephasing



More signal loss.
Moderate T_2^* weighting.
Contrast increases.

Long TE ($\gg T_2^*$)

Strong dephasing



Signal close to zero.
Strong T_2^* weighting.
Image is darker.

Increasing TE $\rightarrow$ stronger T_2^* weighting $\rightarrow$ darker image

TYPICAL T_2^* VALUES AT 1.5T

Tissue	T_2^* (ms)
Liver	15 – 25
Gray Matter	20 – 40
White Matter	25 – 50
Skeletal Muscle	30 – 60
Fat	60 – 100
CSF	100 – 150+
Myocardium	10 – 20

Values vary with field strength, sequence, shimming, and vendor.

CLINICAL RELEVANCE (Uses of T_2^* Contrast)

- ✓ Gradient echo (GRE) sequences are inherently T_2^* weighted.
- ✓ Highly sensitive to susceptibility effects.
- ✓ Useful for detecting:
 - Microbleeds (hemosiderin)
 - Iron deposition (e.g., hemochromatosis)
 - Calcifications
 - Venous structures (SWI)
 - Metal-related artifacts
- ✓ Foundation for SWI (Susceptibility Weighted Imaging) and quantitative susceptibility mapping.

Example: SWI (Brain)



Very sensitive to susceptibility and T_2^* effects.

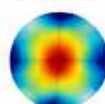
ENGINEERING INSIGHT

- T_2^* is extremely sensitive to shimming quality and system stability.
- Poor shimming $\rightarrow$ broad resonance line $\rightarrow$ short T_2^* $\rightarrow$ rapid signal loss.
- Gradient heating / eddy currents can worsen field inhomogeneity $\rightarrow$ shorter T_2^* .
- Monitor frequency drift, shim values, and gradient performance.

Good Shim



Poor Shim



FIELD SERVICE TIP

- Always perform auto shim before scanning for best T_2^* performance.
- Verify gradient eddy current compensation and pre-emphasis settings.
- Check room and magnet environment (ferromagnetic objects, vibration, HVAC, nearby equipment).
- Metallic implants and susceptibility interfaces will always reduce T_2^* .

COMMON MISCONCEPTION

" T_2^* is the same as T_2 ."

Reality: T_2^* includes both true spin-spin relaxation (T_2) and dephasing from field inhomogeneities. Therefore, T_2^* is always shorter or equal to T_2 .

Better field homogeneity $\rightarrow$ longer T_2^* $\rightarrow$ more signal $\rightarrow$ better image quality.

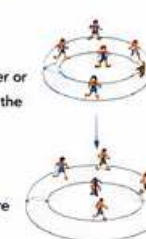
REAL-WORLD ANALOGY

Imagine a group of runners on slightly different circular tracks.

Even if they start together, those on faster or slower tracks lose alignment quickly and the group spreads out faster.

T_2^* is like runners on uneven tracks.

T_2 is like runners on the same track where they only drift apart slowly.



KEY TAKEAWAY

- T_2^* decay is caused by spin-spin interactions AND field inhomogeneities (ΔB_0).
- $T_2^* \leq T_2$ (always shorter or equal).
- Shorter TE $\rightarrow$ less decay $\rightarrow$ brighter image.
- Longer TE $\rightarrow$ more decay $\rightarrow$ darker image.
- T_2^* contrast is powerful for detecting susceptibility-related pathology.

